Abdelrahman Khalifa

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Summary

Computer and Systems Engineering graduate who works on Linux every day: built custom Linux images with Yocto and Buildroot, wrote a Linux kernel module that reports per-process memory and CPU time, set up bridged virtual networks with TFTP boot for test machines, and automated environment setup and CI with Bash. Troubleshoots crashes down to the C code.

Experience

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Built the Node.js/Express + MongoDB REST backend for remote operation and firmware updates, and set up GitHub Actions CI.

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Worked with the Linux printing stack (the CUPS print server, IPP, Avahi/mDNS service discovery) and ran the backend as a separate system service in `cosmic-settings-daemon`, so a crash cannot take down the Settings app.
- Built CI on Ubuntu runners (x86_64 and aarch64) that compiles libcups3 from source, starts the CUPS and Avahi services and runs the tests against them.
- Investigated and fixed crashes and a leaked client in libcups (C); **11 PRs merged upstream** (list).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS Mar 2025

- Implemented the `IP_MULTICAST_LOOP` socket option in its IPv4 networking stack, merged (#86977).

Projects

Virtual test network | *Linux, QEMU, C++* GitHub

- Linux bridge with TAP interfaces, IP forwarding and a TFTP server to network-boot two ARM virtual machines with U-Boot; static IP addressing and a TCP client–server between them and a Raspberry Pi.

Kernel Monitor | *C, Linux kernel* GitHub

- Character device that reports every process's memory, threads, context switches and CPU time; a user-space tool reads it every 2 seconds.

Environment setup automation | *Bash* GitHub

- Idempotent script that installs packages, links configuration files with automatic backups and sets up tools; safe to re-run, with flags to skip steps.

Technical Skills

Linux: Ubuntu/Debian, Arch-based (CachyOS); command line, package management, services, logs, kernel modules, Yocto, Buildroot, QEMU
Networking: TCP/IP, sockets, IPv4 multicast, DNS-SD/mDNS, Linux bridges and TAP interfaces, IP forwarding, TFTP
Scripting & data: Bash, Python, SQL (SQL Server, DuckDB), MongoDB

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors (84.6%). Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.